

・Σ計算①

p, q を k に無関係な定数とする

$$\sum_{k=1}^n (pa_k + qb_k) = \left(\sum_{k=1}^n pa_k + \sum_{k=1}^n qb_k \right) \quad \leftarrow \text{分配の法則}$$

$$= p \sum_{k=1}^n a_k + q \sum_{k=1}^n b_k \quad \leftarrow k \text{ に無関係な定数は } \Sigma \text{ の外へ}$$

(証明)

$$\begin{aligned} & \sum_{k=1}^n (pa_k + qb_k) \\ &= (pa_1 + qb_1) + (pa_2 + qb_2) + \cdots + (pa_n + qb_n) \\ &= p(\underbrace{a_1 + a_2 + \cdots + a_n}_{\sum_{k=1}^n a_k}) + q(\underbrace{b_1 + b_2 + \cdots + b_n}_{\sum_{k=1}^n b_k}) \\ &= p \sum_{k=1}^n a_k + q \sum_{k=1}^n b_k \quad \square \end{aligned}$$

Σ計算のポイント

- I 公式を便える形にハラス
- II 公式を使う
- III 分数でくくって整数係数に
- IV 共通因数でくくる
- V $n=1, 2, \dots$ を代入して検算

(例)

$$\begin{aligned} (1) \quad \sum_{k=1}^n (2k+1) &= 2 \sum_{k=1}^n k + \sum_{k=1}^n 1 \quad \leftarrow \text{I} \\ &= 2 \cdot \frac{1}{2} n(n+1) + n \quad \leftarrow \text{II} \\ &= n(n+1) + n \\ &= n(n+1+1) \quad \leftarrow \text{IV} \\ &= n(n+2), \end{aligned}$$

$$\begin{aligned} (4) \quad \sum_{k=1}^n k(n-k) &= \sum_{k=1}^n (nk - k^2) \quad \leftarrow k \text{ に無関係な } n \text{ は } \Sigma \text{ の外へ} \\ &= n \sum_{k=1}^n k - \sum_{k=1}^n k^2 \quad \leftarrow \text{I} \\ &= n \cdot \frac{1}{2} n(n+1) - \frac{1}{6} n(n+1)(2n+1) \quad \leftarrow \text{II} \\ &= \frac{1}{6} n(n+1) \{3n - (2n+1)\} \quad \leftarrow \text{III IV} \\ &= \frac{1}{6} n(n+1)n(n+1) \end{aligned}$$

$$\begin{aligned} (2) \quad \sum_{k=1}^n k(k+1) &= \sum_{k=1}^n (k^2 + k) \quad \leftarrow \text{I} \\ &= \sum_{k=1}^n k^2 + \sum_{k=1}^n k \quad \leftarrow \text{I} \\ &= \frac{1}{6} n(n+1)(2n+1) + \frac{1}{2} n(n+1) \quad \leftarrow \text{II} \\ &= \frac{1}{6} n(n+1) \{2n+1 + 3\} \quad \leftarrow \text{III IV} \\ &= \frac{1}{6} n(n+1)(2n+4) \\ &= \frac{1}{3} n(n+1)(n+2), \end{aligned}$$

$$\begin{aligned} (5) \quad \sum_{k=n+1}^{2n} k &= \sum_{k=1}^{2n} k - \sum_{k=1}^n k \quad \leftarrow \sum_{k=n+1}^{2n} a_k = a_{n+1} + a_{n+2} + \cdots + a_{2n} \\ &= \frac{1}{2} 2n(2n+1) - \frac{1}{2} n(n+1) \quad \leftarrow \text{II} \\ &= n(2n+1) - \frac{1}{2} n(n+1) \\ &= \frac{1}{2} n \{2(2n+1) - (n+1)\} \quad \leftarrow \text{III IV} \\ &= \frac{1}{2} n(3n+1), \end{aligned}$$

$$\begin{aligned} &= \underbrace{(a_1 + \cdots + a_n)}_{\sum_{k=1}^n a_k} + \underbrace{(a_{n+1} + \cdots + a_{2n})}_{\sum_{k=n+1}^{2n} a_k} - (a_1 + \cdots + a_n) \end{aligned}$$

(3) $n \geq 2$ のとき

$$\begin{aligned} \sum_{k=1}^{n-1} (3k^2 - k + 1) &= 3 \sum_{k=1}^{n-1} k^2 - \sum_{k=1}^{n-1} k + \sum_{k=1}^{n-1} 1 \quad \leftarrow \text{I} \\ &= 3 \cdot \frac{1}{6} (n-1)n(2n-1) - \frac{1}{2} (n-1)n + (n-1) \quad \leftarrow \text{II}, n \text{ は } n-1 \text{ を入れる} \\ &= \frac{1}{2} (n-1)n(2n-1) - \frac{1}{2} (n-1)n + (n-1) \\ &= \frac{1}{2} (n-1) \{n(2n-1) - n + 2\} \quad \leftarrow \text{III IV} \\ &= \frac{1}{2} (n-1)(2n^2 - 2n + 2) \\ &= (n-1)(n^2 - n + 1), \end{aligned}$$